

Question 1:

Consider an infinitely long transmission line consisting of two concentric cylinders having their axes along the z -axis. The cross-section of the line is shown in Fig. 1, where the z -axis is out of the page. The inner conductor has radius a and carries current I , while the outer conductor has inner radius b and thickness t and carries return current $-I$ in the region $b \leq \rho \leq b + t$.

1. In the inner conductor, assume that the current density is given as

$$\mathbf{J} = c_a \rho^2 \mathbf{a}_z$$

Prove that

$$c_a = \frac{2I}{\pi a^4}$$

2. In the outer conductor, assume that the current density is given as

$$\mathbf{J} = c_b \rho^2 \mathbf{a}_z$$

Prove that

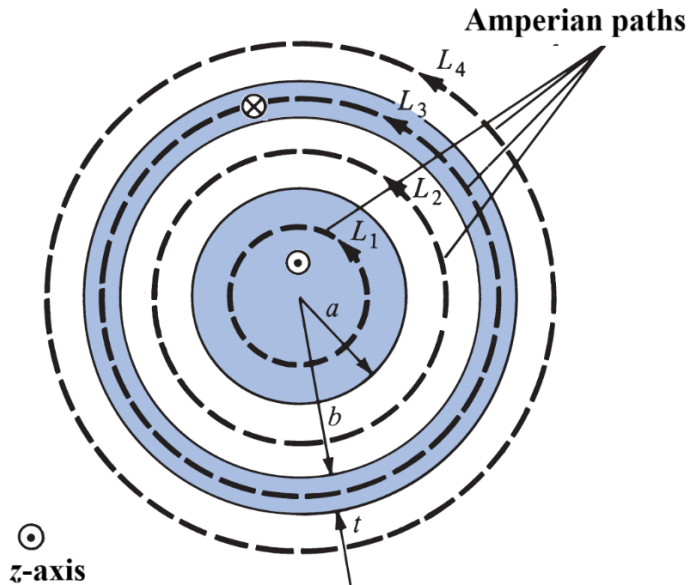
$$c_b = -\frac{2I}{\pi ((b+t)^4 - b^4)}$$

3. Determine magnetic field intensity in the conducting regions. Apply Ampère's law along the Amperian path for the two conducting regions: $0 \leq \rho \leq a$ and $b \leq \rho \leq b + t$.
4. Considering the approximation

$$(1+x)^n \approx 1+nx, \quad \text{for } |x| \ll 1$$

and the fact that $t \ll b$, show that

$$c_b \approx -\frac{I}{2\pi t b^3}$$



Solution:

An infinitely long coaxial transmission line consists of two concentric cylindrical conductors aligned along the z -axis.

- Inner conductor radius: a , carrying current $+I$
- Outer conductor: inner radius b , thickness t , carrying return current $-I$ in $b \leq \rho \leq b + t$
- Cylindrical coordinates (ρ, ϕ, z)

By symmetry, current flows in the z -direction and the magnetic field is circumferential.

1. Inner conductor

The current density is

$$\mathbf{J} = c_a \rho^2 \mathbf{a}_z$$

Total current:

$$I = \iint_S \mathbf{J} \cdot d\mathbf{S}$$

Using $dS = \rho d\rho d\phi$,

$$I = \int_0^{2\pi} \int_0^a c_a \rho^2 (\rho d\rho d\phi) = c_a (2\pi) \int_0^a \rho^3 d\rho$$

$$I = c_a (2\pi) \frac{a^4}{4}$$

Hence,

$$c_a = \frac{2I}{\pi a^4}$$

2. Outer conductor

The current density is

$$\mathbf{J} = c_b \rho^2 \mathbf{a}_z$$

The total current is $-I$:

$$-I = \int_0^{2\pi} \int_b^{b+t} c_b \rho^2 (\rho d\rho d\phi)$$

$$-I = c_b (2\pi) \int_b^{b+t} \rho^3 d\rho = c_b (2\pi) \frac{(b+t)^4 - b^4}{4}$$

Thus,

$$c_b = -\frac{2I}{\pi [(b+t)^4 - b^4]}$$

3. Magnetic field intensity

By symmetry,

$$\mathbf{H} = H_\phi(\rho) \mathbf{a}_\phi$$

Ampère's law:

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} \quad \Rightarrow \quad H_\phi(2\pi\rho) = I_{\text{enc}}$$

Region $0 \leq \rho \leq a$:

$$I_{\text{enc}} = \int_0^{2\pi} \int_0^\rho c_a \rho'^2 (\rho' d\rho' d\phi) = c_a (2\pi) \frac{\rho^4}{4}$$

Substituting c_a :

$$I_{\text{enc}} = I \frac{\rho^4}{a^4}$$

$$\boxed{\mathbf{H}(\rho) = \frac{I\rho^3}{2\pi a^4} \mathbf{a}_\phi, \quad 0 \leq \rho \leq a}$$

Region $b \leq \rho \leq b + t$:

$$I_{\text{enc}} = I + \int_b^\rho c_b \rho'^2 (\rho' d\rho' d\phi)$$

$$I_{\text{enc}} = I + c_b (2\pi) \frac{\rho^4 - b^4}{4}$$

Substituting c_b :

$$I_{\text{enc}} = I \left[1 - \frac{\rho^4 - b^4}{(b+t)^4 - b^4} \right]$$

$$\boxed{\mathbf{H}(\rho) = \frac{I}{2\pi\rho} \left[1 - \frac{\rho^4 - b^4}{(b+t)^4 - b^4} \right] \mathbf{a}_\phi, \quad b \leq \rho \leq b+t}$$

4. Thin outer conductor approximation ($t \ll b$)

$$(b+t)^4 = b^4 \left(1 + \frac{t}{b} \right)^4 \approx b^4 \left(1 + 4\frac{t}{b} \right)$$

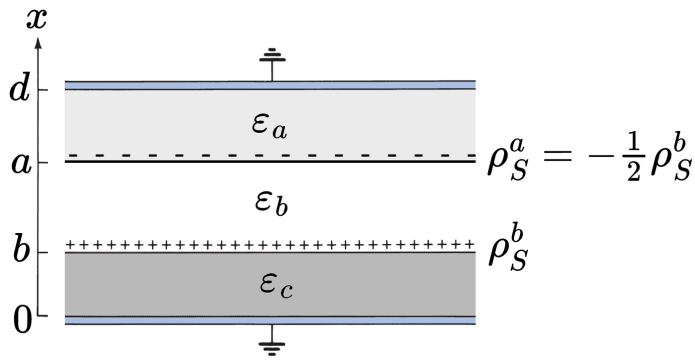
$$(b+t)^4 - b^4 \approx 4b^3 t$$

Substituting into c_b :

$$\boxed{c_b \approx -\frac{I}{2\pi t b^3}}$$

Question 2:

Determine the electric field in each of the three regions within $0 \leq x \leq d$ by considering it an electrostatic boundary-value problem (Laplace's equation). The configuration contains planar interfaces located at $x = 0$, $x = b$, $x = a$, and $x = d$. Surface charge layers are present at the interfaces $x = a$ and $x = b$ as shown in the Figure below:



Solution:

A one-dimensional electrostatic structure with planar interfaces at

$$x = 0, \quad x = b, \quad x = a, \quad x = d$$

containing three dielectric regions:

- Region I: $0 \leq x \leq b$, permittivity ε_c
- Region II: $b \leq x \leq a$, permittivity ε_b
- Region III: $a \leq x \leq d$, permittivity ε_a

Perfectly conducting grounded plates at $x = 0$ and $x = d$.

Surface charge densities:

$$\rho_s^b \text{ at } x = b, \quad \rho_s^a = -\frac{1}{2}\rho_s^b \text{ at } x = a$$

In each dielectric region, no volume charge exists:

$$\nabla^2 V = \frac{d^2 V}{dx^2} = 0$$

Hence,

$$V(x) = Ax + B \quad \Rightarrow \quad \mathbf{E} = -\nabla V = -\frac{dV}{dx} \mathbf{a}_x = \text{constant}$$

Let

$$\mathbf{E} = \begin{cases} E_c \mathbf{a}_x, & 0 \leq x \leq b \\ E_b \mathbf{a}_x, & b \leq x \leq a \\ E_a \mathbf{a}_x, & a \leq x \leq d \end{cases}$$

(i) Conducting plates

$$V(0) = 0, \quad V(d) = 0$$

This implies zero net potential drop:

$$E_c b + E_b(a - b) + E_a(d - a) = 0 \quad (1)$$

(ii) Displacement-field boundary conditions

At $x = b$:

$$\varepsilon_b E_b - \varepsilon_c E_c = \rho_s^b \quad (2)$$

At $x = a$:

$$\varepsilon_a E_a - \varepsilon_b E_b = -\frac{1}{2} \rho_s^b \quad (3)$$

From (2):

$$E_b = \frac{\varepsilon_c}{\varepsilon_b} E_c + \frac{\rho_s^b}{\varepsilon_b} \quad (4)$$

From (3):

$$E_a = \frac{\varepsilon_b}{\varepsilon_a} E_b - \frac{\rho_s^b}{2\varepsilon_a} \quad (5)$$

Substitute (4) into (5):

$$E_a = \frac{\varepsilon_c}{\varepsilon_a} E_c + \frac{\rho_s^b}{\varepsilon_a} - \frac{\rho_s^b}{2\varepsilon_a} = \frac{\varepsilon_c}{\varepsilon_a} E_c + \frac{\rho_s^b}{2\varepsilon_a} \quad (6)$$

Substitute (4) and (6) into (1):

$$E_c b + \left(\frac{\varepsilon_c}{\varepsilon_b} E_c + \frac{\rho_s^b}{\varepsilon_b} \right) (a - b) + \left(\frac{\varepsilon_c}{\varepsilon_a} E_c + \frac{\rho_s^b}{2\varepsilon_a} \right) (d - a) = 0$$

Solving for E_c :

$$E_c = -\rho_s^b \frac{\frac{a-b}{\varepsilon_b} + \frac{d-a}{2\varepsilon_a}}{b + \frac{\varepsilon_c}{\varepsilon_b}(a-b) + \frac{\varepsilon_c}{\varepsilon_a}(d-a)}$$

Then,

$$E_b = \frac{\varepsilon_c}{\varepsilon_b} E_c + \frac{\rho_s^b}{\varepsilon_b}$$

$$E_a = \frac{\varepsilon_c}{\varepsilon_a} E_c + \frac{\rho_s^b}{2\varepsilon_a}$$

$$\mathbf{E}(x) = \begin{cases} E_c \mathbf{a}_x, & 0 \leq x \leq b \\ E_b \mathbf{a}_x, & b \leq x \leq a \\ E_a \mathbf{a}_x, & a \leq x \leq d \end{cases}$$